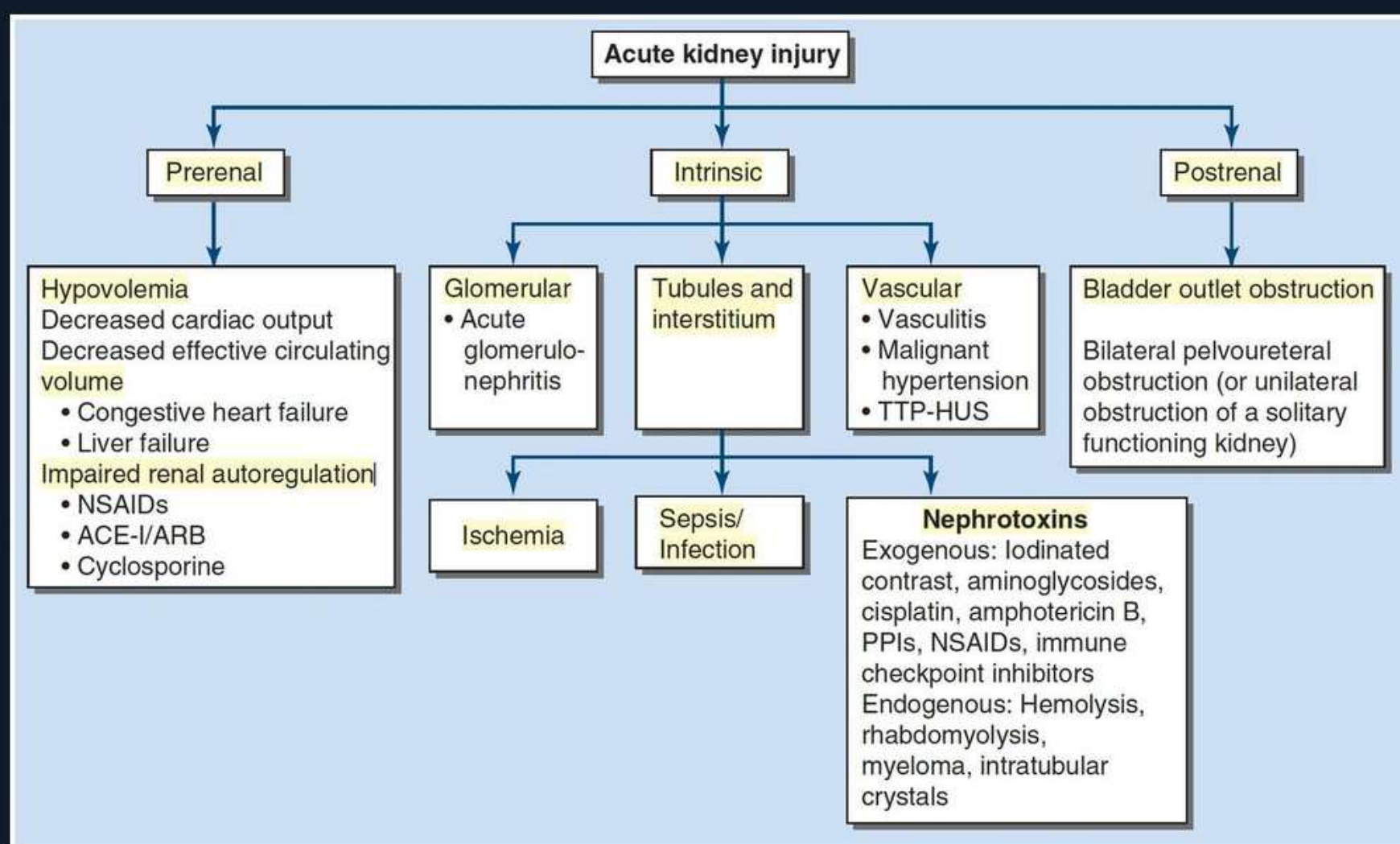


Renal Reference — AKI Algorithm (Harrison)



1. Acute kidney injury

WHAT YOU SEE

Top box: 'Acute kidney injury'

WHAT IT ENCODES

Root node. AKI = abrupt (hours-days) decline in GFR. KDIGO defines it as a rise in serum Cr ≥0.3 mg/dL within 48h, OR ≥1.5× baseline within the prior 7 days, OR urine output <0.5 mL/kg/h for >6h. The algorithm splits AKI into three mechanistic buckets — prerenal (perfusion), intrinsic (parenchymal), and postrenal (obstruction) — because the workup and treatment differ by category.

BOARD TRAP

WRONG answer is diagnosing AKI off a single high creatinine without a baseline. A stable elevated Cr is CKD, not AKI. Discriminator: AKI requires an acute CHANGE — Cr rise ≥0.3 in 48h or ≥1.5× baseline in 7 days; small kidneys on ultrasound point to chronicity instead.

2. Prerenal

WHAT YOU SEE

Left branch box 'Prerenal'

WHAT IT ENCODES

Hypoperfusion of a structurally normal kidney — the single most common cause of AKI (~40-70%). The nephron is intact and conserving: tubules avidly reabsorb Na and water, giving FENa <1%, urine Na <20, urine osm >500, BUN:Cr >20, and bland sediment (hyaline casts only). Fully reversible if perfusion is restored before it progresses to ischemic ATN.

BOARD TRAP

WRONG answer is giving aggressive IV fluids to every prerenal picture. In hepatorenal syndrome and cardiorenal syndrome the 'effective' circulating volume is low but the patient is total-body fluid-OVERLOADED. Discriminator: in HRS treat with albumin + vasoconstrictors (terlipressin/midodrine-octreotide); in cardiorenal AKI diurese/improve cardiac output — fluids worsen both.

3. Intrinsic

WHAT YOU SEE

Middle branch box 'Intrinsic'

WHAT IT ENCODES

Parenchymal injury, subdivided by compartment: glomerular (GN), tubular (ATN), interstitial (AIN), and vascular. ATN is the most common intrinsic cause. Localize by urine sediment: muddy-brown granular casts = ATN; RBC casts/dysmorphic RBCs = GN; WBC casts/eosinophiluria = AIN; bland with systemic vascular signs = vascular.

BOARD TRAP

WRONG answer is lumping all intrinsic AKI as ATN and just supporting it. Missing a rapidly progressive glomerulonephritis (RBC casts, rising Cr) wastes the window for immunosuppression. Discriminator: RBC casts + hematuria + proteinuria mandate urgent nephrology referral, serologies (ANCA, anti-GBM, complement), and often biopsy — not watchful waiting.

4. Postrenal

WHAT YOU SEE

Right branch box 'Postrenal'

WHAT IT ENCODES

Obstruction to urine flow downstream of the kidney — BPH/prostate cancer, bilateral stones, pelvic/retroperitoneal malignancy, neurogenic bladder. Diagnosed by post-void bladder scan and renal ultrasound (hydronephrosis). Relief (Foley, nephrostomy, stent) is often curative; watch for post-obstructive diuresis afterward.

BOARD TRAP

WRONG answer is ruling out obstruction because the ultrasound shows no hydronephrosis. Early or encased (retroperitoneal fibrosis, tumor) obstruction can lack hydronephrosis. Discriminator: high clinical suspicion (anuria, distended bladder, pelvic cancer) warrants CT or retrograde studies even with a 'normal' renal US.

5. Hypovolemia / ↓CO

WHAT YOU SEE

Prerenal list: hypovolemia, ↓cardiac output, ↓effective circulating volume (CHF, liver failure)

WHAT IT ENCODES

Two routes to prerenal: TRUE volume loss (hemorrhage, GI losses, diuretics, burns) and reduced EFFECTIVE arterial volume despite fluid overload (CHF, cirrhosis, nephrotic syndrome, sepsis-induced vasodilation). Both trigger RAAS and ADH, driving Na/water retention — hence FENa <1% and BUN:Cr >20.

BOARD TRAP

WRONG answer is treating a cirrhotic's prerenal azotemia as simple dehydration with crystalloid. Discriminator: hepatorenal syndrome shows FENa <1%, no improvement after 2 days of albumin volume expansion, and no other cause — it needs vasoconstrictors and albumin, not ongoing saline that worsens ascites.

6. Impaired autoregulation

WHAT YOU SEE

Prerenal box lower list: NSAIDs, ACE-I/ARB, cyclosporine

WHAT IT ENCODES

Glomerular pressure is autoregulated by afferent (in) and efferent (out) arteriolar tone. NSAIDs block prostaglandin-mediated afferent VASODILATION → afferent constricts → ↓GFR. ACE-I/ARBs block angiotensin-II-mediated efferent CONSTRICTION → efferent dilates → ↓glomerular pressure → ↓GFR. Calcineurin inhibitors (cyclosporine, tacrolimus) cause afferent vasoconstriction. The triple whammy (NSAID + ACE/ARB + diuretic) is a classic precipitant.

BOARD TRAP

WRONG answer is permanently stopping the ACE-I/ARB after any creatinine bump. A rise of up to ~30% in Cr after starting an ACE-I is expected and acceptable — it reflects reduced intraglomerular pressure and confers long-term renoprotection. Discriminator: stop only if Cr rises >30%, hyperkalemia develops, or bilateral renal artery stenosis is suspected (large jump signals it).

7. Glomerular

WHAT YOU SEE

Intrinsic subbox 'Glomerular: acute glomerulonephritis'

WHAT IT ENCODES

Glomerular inflammation → nephritic syndrome: hematuria with dysmorphic RBCs, RBC casts, subnephrotic proteinuria, HTN, edema. Work up with complement levels and serologies — low complement: post-strep GN, lupus, MPGN, cryoglobulinemia; normal complement (pauci-immune/anti-GBM): ANCA vasculitis (GPA/MPA), anti-GBM/Goodpasture, IgA nephropathy. RPGN = crescents on biopsy; treat urgently with steroids ± cyclophosphamide/rituximab/plasmapheresis.

BOARD TRAP

WRONG answer is calling glomerular hematuria a 'UTI' or stone. Discriminator: RBC CASTS and DYSMORPHIC RBCs are pathognomonic of glomerular bleeding — isomorphic RBCs with clots suggest urologic/lower-tract source. Casts mean send serologies and consider biopsy, not antibiotics.

8. Tubules & interstitium

WHAT YOU SEE

Intrinsic subbox 'Tubules and interstitium'

WHAT IT ENCODES

Two entities: ATN (tubular cell death from ischemia, sepsis, or nephrotoxins → muddy-brown granular casts, FENa >2%, isosthenuric urine osm <350) and AIN (immune/allergic interstitial inflammation → WBC casts, sterile pyuria, eosinophiluria, often after a drug). ATN has classic phases: initiation, maintenance (oliguria), and recovery (polyuria).

BOARD TRAP

WRONG answer is treating ATN as if fluids will fix it. Once tubular necrosis has occurred, a fluid bolus does NOT rapidly reverse it — recovery takes days to weeks of supportive care. Discriminator: prerenal improves promptly with fluids and has FENa <1% with bland urine; established ATN has FENa >2% and muddy-brown casts and will not bounce back with a bolus.

9. Vascular

WHAT YOU SEE

Intrinsic subbox 'Vascular': vasculitis, malignant HTN, TTP-HUS

WHAT IT ENCODES

Large-vessel (renal artery thromboembolism/dissection, renal vein thrombosis) and small-vessel disease (thrombotic microangiopathy: TTP-HUS, malignant HTN, scleroderma renal crisis, atheroembolic disease). TMA = MAHA (schistocytes, ↓haptoglobin, ↑LDH) + thrombocytopenia + AKI. Scleroderma renal crisis → treat with ACE inhibitors (the one place an ACE-I is first-line in AKI).

BOARD TRAP

WRONG answer is giving platelet transfusions in TTP for the low platelets. This can worsen microthrombi. Discriminator: TTP (ADAMTS13 deficiency) is treated with urgent plasma exchange ± steroids/caplacizumab — platelet transfusion is contraindicated except for life-threatening bleeding or before a procedure.

10. Ischemia / Sepsis

WHAT YOU SEE

Boxes 'Ischemia' and 'Sepsis/Infection'

WHAT IT ENCODES

Ischemia (prolonged hypotension, shock, post-surgical) and sepsis are the leading triggers of ATN — prerenal hypoperfusion that crosses the threshold into tubular necrosis. The proximal tubule and outer-medulla thick ascending limb are most vulnerable (high metabolic demand, watershed O2 supply). Sepsis adds inflammatory and microvascular injury.

BOARD TRAP

WRONG answer is assuming early sepsis-associated AKI must have a high FENa like other ATN. In the first hours, sepsis and ischemia can present with a LOW FENa (<1%) before frank tubular necrosis sets in — a low-FENa exception. Discriminator: don't exclude evolving ATN just because FENa is low early; follow the sediment (muddy-brown casts) and the clinical trajectory.

11. Nephrotoxins — exogenous

WHAT YOU SEE

Nephrotoxins box top: iodinated contrast, aminoglycosides, cisplatin, amphotericin B, PPIs, NSAIDs, immune checkpoint inhibitors

WHAT IT ENCODES

Classic exogenous tubular/interstitial toxins. ATN pattern: aminoglycosides (nonoliguric, ~day 5-7, dose/duration related), cisplatin (Mg wasting), amphotericin B (distal RTA, K/Mg wasting), iodinated contrast (Cr rises 24-48h, peaks 3-5d, recovers by 7d, low FENa). AIN pattern: PPIs, NSAIDs (also cause membranous/minimal change), checkpoint inhibitors, beta-lactams.

BOARD TRAP

WRONG answer is reflexively giving N-acetylcysteine and bicarbonate to 'prevent' contrast nephropathy. Large trials (PRESERVE) showed no benefit over isotonic saline. Discriminator: the evidence-based prophylaxis is IV isotonic saline volume expansion and minimizing contrast dose — NAC/bicarbonate are not recommended.

12. Nephrotoxins — endogenous

WHAT YOU SEE

Nephrotoxins box bottom: hemolysis, rhabdomyolysis, myeloma, intratubular crystals

WHAT IT ENCODES

Endogenous pigments/proteins/crystals: rhabdomyolysis (myoglobin → ↑CK >5000, heme-positive dipstick with NO RBCs, hyperK, hyperphosphatemia, hypocalcemia), hemolysis (hemoglobin), myeloma cast nephropathy (light chains; suspect with high anion-gap, anemia, hypercalcemia, age >60), and crystal nephropathies (tumor lysis uric acid, acyclovir, methotrexate, ethylene glycol oxalate). Pigment nephropathies are a low-FENa exception.

BOARD TRAP

WRONG answer is trusting a 'positive blood' dipstick with no RBCs on microscopy as hematuria. Discriminator: heme-positive dipstick + absent RBCs = myoglobinuria (rhabdo) or hemoglobinuria — check CK and a peripheral smear/haptoglobin; treat rhabdo with aggressive IV fluids targeting brisk urine output.

13. Bladder outlet / pelvic obstruction

WHAT YOU SEE

Postrenal box: bladder outlet obstruction, bilateral pelvocalyceal obstruction (or unilateral in solitary kidney)

WHAT IT ENCODES

For obstruction to cause AKI you need BILATERAL involvement (or obstruction of a single functioning kidney) because one healthy kidney can maintain GFR. Most common is bladder outlet obstruction (BPH, prostate cancer, neurogenic bladder); upper-tract causes are bilateral stones, pelvic/retroperitoneal tumor, or retroperitoneal fibrosis. Relief can trigger post-obstructive diuresis — monitor volume and electrolytes.

BOARD TRAP

Unilateral ureteral obstruction with a normal contralateral kidney causes AKI — this is the **WRONG** answer. A single normal kidney maintains overall GFR, so unilateral obstruction does NOT raise creatinine (though it may cause pain/hydronephrosis). Discriminator: postrenal AKI requires bilateral obstruction or obstruction of a solitary functioning kidney.
